Charles (Tzu-Wei) Lee, Ph.D. Candidate

Neuroimaging & MRI Physics | Multimodal ML | LLM Agents & Inference Systems | Dynamical Systems
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SUMMARY

Ph.D. candidate in Biomedical Engineering at Vanderbilt University with 8+ years in biotechnology and neuroscience and 7+ years of machine learning research. Work spans MRI physics (pulse sequence programming and image reconstruction at 3T, 7T, and 15.2T), multimodal representation learning on large-scale neuroimaging and genomics data, and LLM agent systems for clinical decision support. Authorized to work in the U.S. (STEM OPT; EB-2 NIW pending).

EDUCATION

Vanderbilt University, Nashville, TN — Aug. 2021 – Present
Ph.D. in Bioengineering and Biomedical Engineering, Expected Dec. 2026 ~ 2027

- Advisor: Prof. John C. Gore.
- Committee: Prof. Catie Chang, Prof. Adam Anderson, Prof. Kurt Schilling, Prof. Chen Li Min
- Dissertation: Beyond fMRI Assumptions: Acquisition, Decomposition, and the Biophysical Origin of the BOLD Signal Beyond Gray Matter

University of Arizona, Tucson, AZ — Jan. 2019 – Feb. 2019
Visiting Scholar | Advisor: Prof. Nan-Kuei Chen

- Resolved and assessed artifacts in MRI reconstruction – phase imaging.

Taipei Medical University, Taipei, Taiwan — Sept. 2016 – June 2020
B.S. in Biomedical Engineering | Last-60-credit GPA: 3.9/4.0

RESEARCH EXPERIENCE

Graduate Research Assistant, Vanderbilt University Institute of Imaging Science — Aug. 2021 – Present
MRI Physics and Neuroimaging

- **In-vivo mouse spinal cord fMRI at 15.2T:** Built the first end-to-end pipeline, covering experimental design, C++ pulse sequences, 3D-printed hardware for sub-3 mm anatomical stabilization, and automated Python/MATLAB preprocessing and statistics that overcame severe motion and low SNR. (ISMRM 2025 Oral; ISMRM 2026 Oral; paper under review)
- **Multi-shot rosette DWI:** Implemented a multi-shot rosette readout sequence and co-developed its image reconstruction on 7T and 15.2T scanners, correcting B0 inhomogeneity, eddy currents, and phase variations to remove signal attenuation from multi-shot averaging. (SSRN paper)
- **BOLD & Hemodynamic Response Function modeling:** Developed time- and frequency-domain (FFT/spectral) models of the hemodynamic response, using dynamical systems that link neuronal input, CBF, CBV, and $CMRO_2$ to the BOLD signal, to identify region- and condition-specific effects. (2 papers under review, [2]; 2 more manuscripts in preparation)

Clinical AI and Multimodal Learning

- **Multimodal learning and white-matter brain charts:** Built a Spatiotemporal Masked Autoencoder mapping fMRI and genomics into a shared latent space for disease-signature discovery (INDI, HCP, UK Biobank; 600+ TB, 60,000+ subjects), and lifespan normative charts of 8 white-matter functional measures from 9,948 participants (ages 5–96), showing that WM–WM FC peaks at age 24, with calibrated & reproducible centiles. (1 manuscript)
- **Primate behavior modeling:** Built a deep learning motion-tracking system for non-human primate palm movements; cross-condition benchmarks and error analysis guided preprocessing and model changes that enabled quantitative motor-behavior analysis for the lab's publications.

- **CopathAgent:** Designed a clinical decision support framework for multimodal dementia profiling, in which a multi-agent system delivers ranked, auditable co-pathology attributions for downstream diagnostic workflows. (Manuscript in preparation)
- **TissueLoom, tissue-aware connectomics in Alzheimer's disease:** Developed a tissue-aware model of GM–GM, WM–WM and GM–WM functional connectivity and applied it to 529 participants from preclinical AD to dementia, with ablation, coupling and AD-biomarker analyses. GM–WM connectivity was the most informative during the preclinical stage in cingulum bundles and default-mode regions; altered couplings rose from 0 at preclinical AD to 75 at dementia, and the predictions tracked amyloid and tau. (ISMRM 2027 submitted, 1 manuscript)

Undergraduate Research Assistant, Taipei Medical University Hospital May 2018 – Aug. 2020

- Implemented MRI protocols for a 20-subject Parkinson's disease study and performed quantitative SWI/QSM analysis, improving contrast-to-noise ratio for nigrosome-1 visualization. (ISMRM 2020; Tomography 2021)
- Contributed to a radiogenomic analysis of perfusion MRI in glioblastoma and a study of APOE-ε4 effects on cholinergic circuitry. (ISMRM 2020; Frontiers in Neurology 2021)

Research Assistant, Applied Mechanics, National Taiwan University May 2017 – July 2018

- Designed and prototyped microchannel chips for electricity-free blood centrifugation; performed ELISA protein analysis, 3D modeling, and silicon-wafer microfabrication.

WORK EXPERIENCE

Clinical Research Assistant, Best-Image & Siemens Healthineers, Taiwan Sept. 2020 – Jan. 2021

- Programmed an ASL imaging sequence in Siemens IDEA (VE11C) on a 3T PRISMA scanner.
- Built Python ML evaluation pipelines for neuroimaging, including MRI denoising and segmentation, to improve image quality for clinical diagnosis.
- Brought a hospital's newly installed MRI system into research use by optimizing acquisition protocols and validating data quality, which supported the launch of 3 clinical and translational research projects.

Engineering Intern, Dynamic Medical Technologies Inc. July 2019 – Sept. 2019

- Installed, maintained, and troubleshot laser instruments used in plastic surgery.

TEACHING EXPERIENCE

Teaching Assistant, Taipei Medical University

- MATLAB Programming & Application (Fall 2020): ~20 students, bilingual (Mandarin/English)
- Medical Imaging & Neuroscience (Fall 2019, Fall 2020): ~30 students, (Mandarin/English)
- Medical Imaging Systems (Fall 2019): ~15 people, Mandarin & English
- Neuroscience & Molecular Imaging (Fall 2019): ~10 people, English
- Advanced Medical Imaging Approach (Spring 2019): ~10 people, English

SKILLS

Programming: Python, C++, C#, MATLAB, Julia, R, SQL, Go, Node.js, HTML, Bash
ML & LLM: PyTorch, JAX/XLA, TensorFlow, scikit-learn, Hugging Face, LangChain, vLLM, SGLang, Triton, CUDA
MRI & Signal Processing: Pulse sequence programming (Siemens IDEA, Bruker, C++), image reconstruction, DWI, ASL, SWI/QSM, fMRI preprocessing, time- and frequency-domain analysis, dynamical systems modeling
Infrastructure & Cloud: Docker, Kubernetes, Ray, AWS (S3, Lambda, IAM, Glue, SageMaker), GCP (Vertex AI), Supabase, Redis, Linux, Git
Laboratory: In-vivo animal imaging, hardware prototyping (3D printing), ELISA, microfabrication